Geometric Methods of Mathematical Physics

Exercises.

Chapter 2

## Exercise 2.1

$$\hat{r} = \cos\theta\,\hat{x} + \sin\theta\,\hat{y}, \qquad \hat{\theta} = -\sin\theta\,\hat{x} + \cos\theta\,\hat{y},$$

with $\hat{x} = \partial/\partial x$, $\hat{y} = \partial/\partial y$.

Show $\hat{r}, \hat{\theta}$ are non-coordinate basis $\Rightarrow$ show $[\hat{r}, \hat{\theta}] \neq 0$.

$$\begin{aligned}[\hat{r}, \hat{\theta}] &= \hat{r}\hat{\theta} - \hat{\theta}\hat{r} \\ &= (c\hat{x} + s\hat{y})(-s\hat{x} + c\hat{y}) - (-s\hat{x} + c\hat{y})(c\hat{x} + s\hat{y}),\end{aligned}$$

where $c = \cos\theta$, $s = \sin\theta$.

$$\begin{aligned}[\hat{r}, \hat{\theta}] &= -c\hat{x}s\hat{x} + c\hat{x}c\hat{y} - s\hat{y}s\hat{x} + s\hat{y}c\hat{y} \\ &\quad - (-s\hat{x}c\hat{x} - s\hat{x}s\hat{y} + c\hat{y}c\hat{x} + c\hat{y}s\hat{y}) \\ &= [s\hat{x}, c\hat{x}] + [c\hat{x}, c\hat{y}] + [s\hat{x}, s\hat{y}] + [s\hat{y}, c\hat{y}].\end{aligned}$$

$$c\hat{x}s\hat{x} = cs\hat{x}\hat{x} + c\hat{x}(s)\hat{x}.$$

$$\begin{aligned}[s\hat{x}, c\hat{x}] &= sc\hat{x}\hat{x} + s\hat{x}(c)\hat{x} - cs\hat{x}\hat{x} - c\hat{x}(s)\hat{x} \\ &= s\hat{x}(c)\hat{x} - c\hat{x}(s)\hat{x}, \qquad \text{use chain rule} \\ &= s\frac{\partial\theta}{\partial x}\frac{\partial c}{\partial\theta}\hat{x} - c\frac{\partial\theta}{\partial x}\frac{\partial s}{\partial\theta}\hat{x} \\ &= -s^2\frac{\partial\theta}{\partial x}\hat{x} - c^2\frac{\partial\theta}{\partial x}\hat{x} = -\frac{\partial\theta}{\partial x}\hat{x},\end{aligned}$$

with $\partial s/\partial\theta = c$, $\partial c/\partial\theta = -s$, $s^2 + c^2 = 1$.

$$\Rightarrow \quad [s\hat{x}, c\hat{x}] = -\frac{\partial\theta}{\partial x}\hat{x}.$$

Similarly,

$$\begin{aligned}[c\hat{x}, c\hat{y}] &= c\hat{x}(c)\hat{y} - c\hat{y}(c)\hat{x} \\ &= c\frac{\partial\theta}{\partial x}\frac{\partial c}{\partial\theta}\hat{y} - c\frac{\partial\theta}{\partial y}\frac{\partial c}{\partial\theta}\hat{x} \\ &= cs\left(\frac{\partial\theta}{\partial y}\hat{x} - \frac{\partial\theta}{\partial x}\hat{y}\right), \\ [s\hat{x}, s\hat{y}] &= s\hat{x}(s)\hat{y} - s\hat{y}(s)\hat{x} \\ &= s\left(\frac{\partial\theta}{\partial x}\frac{\partial s}{\partial\theta}\hat{y} - \frac{\partial\theta}{\partial y}\frac{\partial s}{\partial\theta}\hat{x}\right) \\ &= -cs\left(\frac{\partial\theta}{\partial y}\hat{x} - \frac{\partial\theta}{\partial x}\hat{y}\right), \\ [s\hat{y}, c\hat{y}] &= s\hat{y}(c)\hat{y} - c\hat{y}(s)\hat{y} \\ &= s\frac{\partial\theta}{\partial y}\frac{\partial c}{\partial\theta}\hat{y} - c\frac{\partial\theta}{\partial y}\frac{\partial s}{\partial\theta}\hat{y} \\ &= (-s^2 - c^2)\frac{\partial\theta}{\partial y}\hat{y} = -\frac{\partial\theta}{\partial y}\hat{y}.\end{aligned}$$

Collecting the results we get

$$[\hat{r},\hat{\theta}] = -\frac{\partial\theta}{\partial x}\hat{x} - \frac{\partial\theta}{\partial y}\hat{y} \neq 0 \quad \text{in general.}$$

$\Rightarrow$ Since $[\hat{r},\hat{\theta}]$ doesn't vanish in general it is not a coordinate basis. $\square$

## Exercise 2.2

### a)

Use (2.6) to prove (2.12).

$$(2.6): \quad x^i(\lambda_0+\varepsilon) = \exp\left[\varepsilon\frac{d}{d\lambda}\right] x^i\Big|_{\lambda_0},$$
$$(2.12): \quad x^i(B) - x^i(A) = \varepsilon^2[\vec{V},\vec{W}] + O(\varepsilon^3).$$

Situation:

$$\begin{aligned} x(B) &= x(Q+\varepsilon), && \text{along } d/d\lambda,\\ x(Q) &= x(P+\varepsilon), && \text{along } d/d\mu,\\ x(A) &= x(R+\varepsilon), && \text{along } d/d\mu,\\ x(R) &= x(P+\varepsilon), && \text{along } d/d\lambda. \end{aligned}$$

$$\begin{aligned} x^i(B) - x^i(A) &= x^i(Q+\varepsilon) - x^i(R+\varepsilon), \qquad \text{use 2.6}\\ &= \exp\left[\varepsilon\frac{d}{d\lambda}\right] x^i\Big|_Q - \exp\left[\varepsilon\frac{d}{d\mu}\right] x^i\Big|_R\\ &= \exp\left(\varepsilon\frac{d}{d\lambda}\right) x^i(P+\varepsilon) - \exp\left[\varepsilon\frac{d}{d\mu}\right] x^i(P+\varepsilon)\\ &= \exp\left(\varepsilon\frac{d}{d\lambda}\right)\exp\left(\varepsilon\frac{d}{d\mu}\right) x^i\Big|_P - \exp\left[\varepsilon\frac{d}{d\mu}\right]\exp\left(\varepsilon\frac{d}{d\lambda}\right) x^i\Big|_P\\ &= \left[\exp(\varepsilon\, d/d\lambda), \exp(\varepsilon\, d/d\mu)\right] x^i\big|_P\\ &= \varepsilon^2\left[\frac{d}{d\lambda},\frac{d}{d\mu}\right] x^i\Big|_P + O(\varepsilon^3)\\ &= \varepsilon^2[\vec{V},\vec{W}] + O(\varepsilon^3). \end{aligned}$$

Here $x^i_Q = x^i(Q) = x^i(P+\varepsilon)$. Note that the two $x^i(P+\varepsilon)$ are different.

$$\vec{V} = \frac{d}{d\lambda}x^i\Big|_P, \qquad \vec{W} = \frac{d}{d\mu}x^i\Big|_P.$$

### b)

Prove that

$$\exp\left[a\frac{d}{d\lambda} + b\frac{d}{d\mu}\right] = \exp[a\, d/d\lambda]\exp[b\, d/d\mu] \tag{1}$$

for all $a$ & $b$ iff $[d/d\lambda, d/d\mu] = 0$.

Summation is commutative, i.e.,

$$a\frac{d}{d\lambda} + b\frac{d}{d\mu} = b\frac{d}{d\mu} + a\frac{d}{d\lambda}.$$

(But multiplication of vectors isn't necessarily.)

$$[\exp[a\,d/d\lambda], \exp[b\,d/d\mu]] = \exp(a\,d/d\lambda)\exp(b\,d/d\mu) - \exp(b\,d/d\mu)\exp(a\,d/d\lambda).$$

Assuming (1) holds:

$$[\exp(a\,d/d\lambda), \exp(b\,d/d\mu)] = \exp\left(a\frac{d}{d\lambda} + b\frac{d}{d\mu}\right) - \exp\left(b\frac{d}{d\mu} + a\frac{d}{d\lambda}\right) = 0.$$

So (1) working implies $[d/d\lambda, d/d\mu] = 0$, but at the same time if (1) works but we assume $[d/d\lambda, d/d\mu] \neq 0$ then we get a contradiction.

$$\Rightarrow \quad \exp\left[a\frac{d}{d\lambda} + b\frac{d}{d\mu}\right] = \exp\left[a\frac{d}{d\lambda}\right]\exp\left[b\frac{d}{d\mu}\right] \quad \text{iff} \quad \left[\frac{d}{d\lambda}, \frac{d}{d\mu}\right] = 0.$$

Another way to see this is to expand the exponential of (1) as Taylor series & compare both sides.

## Exercise 2.3

Prove that any 3 $C^\infty$ vector fields $\vec{X}, \vec{Y}, \vec{Z}$ satisfy the Jacobi identity:

$$[[\vec{X}, \vec{Y}], \vec{Z}] + [[\vec{Y}, \vec{Z}], \vec{X}] + [[\vec{Z}, \vec{X}], \vec{Y}] = 0.$$

This is quite trivial:

$$\begin{aligned}
0 &= [\vec{X}, \vec{Y}]\vec{Z} - \vec{Z}[\vec{X}, \vec{Y}] + [\vec{Y}, \vec{Z}]\vec{X} - \vec{X}[\vec{Y}, \vec{Z}] + [\vec{Z}, \vec{X}]\vec{Y} - \vec{Y}[\vec{Z}, \vec{X}] \\
&= [\vec{X}\vec{Y} - \vec{Y}\vec{X}, \vec{Z}] + [\vec{Y}\vec{Z} - \vec{Z}\vec{Y}, \vec{X}] + [\vec{Z}\vec{X} - \vec{X}\vec{Z}, \vec{Y}] \\
&= \vec{X}\vec{Y}\vec{Z} - \vec{Y}\vec{X}\vec{Z} - \vec{Z}\vec{X}\vec{Y} + \vec{Z}\vec{Y}\vec{X} \\
&\quad + \vec{Y}\vec{Z}\vec{X} - \vec{Z}\vec{Y}\vec{X} - \vec{X}\vec{Y}\vec{Z} + \vec{X}\vec{Z}\vec{Y} \\
&\quad + \vec{Z}\vec{X}\vec{Y} - \vec{X}\vec{Z}\vec{Y} - \vec{Y}\vec{Z}\vec{X} + \vec{Y}\vec{X}\vec{Z} \\
&= 0,
\end{aligned}$$

as all terms cancel. $\square \Rightarrow$ Jacobi identity holds.

## Exercise 2.4

A matrix is $\binom{1}{1}$-tensor so a linear transformation $T$ is a map

$$T\left(\binom{1}{1}\text{-tensor}\right) \longrightarrow \binom{1}{1}\text{-tensor},$$

i.e., it spits out one one-form and one vector when given as arguments one one-form and one vector. A $\binom{1}{1}$ tensor is a map that takes one one-form & one vector & gives real number. So we see that when $T$ is given 2 one-forms & 2 vectors as arguments it gives real number as a value. I.e., it is a linear function which takes as arguments 2 one-forms & 2 vectors and whose value is a real number. This is a definition of $\binom{2}{2}$-tensor $\Rightarrow T$ is $\binom{2}{2}$ tensor. $\square$

## Exercise 2.5

### a)

A general $\binom{2}{0}$ tensor has $n^2$ independent components meanwhile a vector has $n$ components $\Rightarrow$ Outer product of two vectors has $2n$ components of which only $2n-1$ are independent because of rescaling $aV \otimes \frac{1}{a}W$ for $T = V \otimes W$, $V, W$ are vectors, because this fixes one parameter.

So we have $n^2$ for general $\binom{2}{0}$ tensor & $2n-1$ for outer product of two vectors and as such these vectors' outer product can't describe a general $\binom{2}{0}$ tensor.

**b)**

$\vec{V}, \tilde{\omega}, \vec{V} \otimes \tilde{\omega}$ components are

$$\begin{aligned} \vec{V} \otimes \tilde{\omega} &= V^i \frac{\partial}{\partial x^i} \otimes \omega^j \, dx_j \\ &= \underbrace{V^i \omega^j}_{\text{Component}} \quad \underbrace{\left( \frac{\partial}{\partial x^i} \otimes dx_j \right)}_{\text{basis vector outer product}} . \end{aligned}$$

## Exercise 2.6

Prove: Set of all $\binom{2}{0}$ tensors at $P$ is a vector space under addition defined by analogy with

$$(\tilde{\omega} + \tilde{\sigma})(\vec{V}) = \tilde{\omega}(\vec{V}) + \tilde{\sigma}(\vec{V}).$$

The analogous equation for $T, E$ $\binom{2}{0}$ tensors would be:

$$(T + E)(\tilde{\omega}, \tilde{\sigma}) = T(\tilde{\omega}, \tilde{\sigma}) + E(\tilde{\omega}, \tilde{\sigma}).$$

Since $T \in \mathbb{R}^n \times \mathbb{R}^n$, $E \in \mathbb{R}^n \times \mathbb{R}^n$, then $T + E \in \mathbb{R}^n \times \mathbb{R}^n$. And the multiplication by scalar is defined as like in vector case:

$$a(T + E) = aT + aE, \qquad (a + b)T = aT + bT, \qquad abT = a(bT), \quad T = 1 \cdot T.$$

$\Rightarrow$ A set of all $\binom{2}{0}$ tensors at $P$ is a vector space.

Now to calculate the basis we use similar technique as in Eq. (2.24)–(2.26):

$$\begin{aligned} T(\tilde{\omega}, \tilde{\sigma}) &= T(\omega_i \tilde{\omega}^i, \sigma_j \tilde{\omega}^j) = \omega_i \sigma_j T(\tilde{\omega}^i, \tilde{\omega}^j), \\ T(\tilde{\omega}, \tilde{\sigma}) &= \vec{e}_i(\tilde{\omega}) \vec{e}_j(\tilde{\sigma}) T(\tilde{\omega}^i, \tilde{\omega}^j), \end{aligned}$$

where $\tilde{\omega}^i$ is the basis of one-forms. A $\binom{2}{0}$ tensor maps 2 one-forms to real numbers.

$T(\tilde{\omega}^i, \tilde{\omega}^j)$ are the components of $T$.

$$\Rightarrow \quad T(\tilde{\omega}, \tilde{\sigma}) = T^{ij} \vec{e}_i(\tilde{\omega}) \vec{e}_j(\tilde{\sigma}), \qquad \tilde{\omega}, \tilde{\sigma} \text{ are arbitrary.}$$

$$\Rightarrow \quad T = T^{ij} \vec{e}_i \otimes \vec{e}_j. \quad \square$$

The basis of $\binom{2}{0}$ tensors is $\vec{e}_i \otimes \vec{e}_j$, i.e., the set of all $\binom{2}{0}$ tensors is $T_P \otimes T_P$ space.

## Exercise 2.7

$Q^{ijk}{}_{\ell m}$ has 2 contractions for each upstairs indices so $3 \cdot 2 = 6$ $\binom{2}{1}$ tensors may be made. AFTER this we have tensors of form $Q^{ijk}{}_{im}$.

The second contraction can be done $2 \cdot 1 = 2$ times from $\binom{2}{1}$ tensor, meaning from initial $\binom{3}{2}$ tensor we can make $6 \cdot 2 = 12$ $\binom{1}{0}$ tensors by two contractions.

## Exercise 2.8

$A, B$ are $\binom{1}{1}$ tensors, i.e., vector-valued linear functions of vectors. We define

$$C(\vec{V}) \equiv B(A(\vec{V})).$$

This is a $\binom{1}{1}$ tensor because $A(\vec{V}) = \vec{W}$ is a vector, meaning

$$C(\vec{V}) = B(\vec{W}),$$

and $B$ is $\binom{1}{1}$ tensor, i.e., vector-valued linear function of vectors.

$$A(\vec{V}) = A^i{}_j V^j \vec{e}_i = A^i{}_j \tilde{\omega}^j(\vec{V})\vec{e}_i, \qquad A^i{}_j \text{ is the component.}$$
$$\Rightarrow \quad A = A^i{}_j \vec{e}_i \otimes \tilde{\omega}^j, \qquad \text{similarly} \quad B = B^i{}_j \vec{e}_i \otimes \tilde{\omega}^j.$$

$$\begin{aligned} C(\vec{V}) &= B(A(\vec{V})) = B\big(A^i{}_j \tilde{\omega}^j(\vec{V})\vec{e}_i\big) \\ &= A^i{}_j \tilde{\omega}^j(\vec{V}) B(\vec{e}_i) \\ &= A^i{}_j \tilde{\omega}^j(\vec{V}) B^k{}_\ell \underbrace{\tilde{\omega}^\ell(\vec{e}_i)}_{=\delta^\ell_i} \vec{e}_k, \\ C(\vec{V}) &= A^i{}_j B^k{}_i \tilde{\omega}^j(\vec{V})\vec{e}_k, \end{aligned}$$

this is independent of $\vec{V}$.

$$\Rightarrow \quad C = A^i{}_j B^k{}_i \tilde{\omega}^j \otimes \vec{e}_k \quad \Rightarrow \quad C^i{}_j = B^i{}_k A^k{}_j, \qquad \text{(Lil bit of renaming).}$$

This corresponds to component of matrix product when comparing with the linear transformation defined in §1.6:

$$(BA)_{ij} = \sum_k B_{ik} A_{kj}. \tag{1.32}$$

## Exercise 2.9

Show that $\binom{2}{0}$ tensor's components transform as two vectors, i.e.,

$$T^{i'j'} = \Lambda^{i'}{}_k \Lambda^{j'}{}_\ell T^{k\ell}.$$
$$T = T^{ij}\vec{e}_i \otimes \vec{e}_j \qquad \text{for } \binom{2}{0}\ T.$$

In the new basis $\vec{e}_{i'} = \Lambda^j{}_{i'}\vec{e}_j$.

$$\begin{aligned} T &= T^{i'j'}\vec{e}_{i'} \otimes \vec{e}_{j'} \\ &= T^{i'j'}\big(\Lambda^i{}_{i'}\vec{e}_i \otimes \Lambda^j{}_{j'}\vec{e}_j\big) \\ &= T^{i'j'}\Lambda^i{}_{i'}\Lambda^j{}_{j'}\vec{e}_i \otimes \vec{e}_j = T^{ij}\vec{e}_i \otimes \vec{e}_j. \\ &\Rightarrow \quad T^{i'j'}\Lambda^i{}_{i'}\Lambda^j{}_{j'} = T^{ij}. \\ \Rightarrow \quad T^{i'j'} &= \Lambda^{i'}{}_i \Lambda^{j'}{}_j T^{ij} = \Lambda^{i'}{}_k \Lambda^{j'}{}_\ell T^{k\ell}. \quad \square \end{aligned}$$

For $\binom{N}{N'}$ tensor we have

$$T = T^{ij\cdots N}{}_{\ell m\cdots N'}\, \vec{e}_i \otimes \vec{e}_j \otimes \cdots \otimes \vec{e}_N \otimes \tilde{\omega}^\ell \otimes \cdots \otimes \tilde{\omega}^{N'}.$$

Each basis vector transforms as

$$\vec{e}_{i'} = \Lambda^j{}_{i'}\vec{e}_j, \quad \text{giving} \quad \vec{e}_j = \Lambda^{i'}{}_j \vec{e}_{i'}.$$

Each basis one-form transforms as

$$\tilde{\omega}^{j'} = \Lambda^{j'}{}_k \tilde{\omega}^k, \quad \text{giving} \quad \tilde{\omega}^k = \Lambda^k{}_{i'}\tilde{\omega}^{i'}.$$

Thus under basis transformation we get

$$T^{i'j'k'\cdots N^*}{}_{\ell' m'\cdots N^{*(1)}} = T^{ijk\cdots N}{}_{\ell m\cdots N'}\, \Lambda^{i'}{}_i \Lambda^{j'}{}_j \cdots \Lambda^{N^*}{}_N \Lambda^\ell{}_{\ell'} \Lambda^m{}_{m'} \cdots \Lambda^{N'}{}_{N^{*(1)}}.$$

Each basis vector or one-form contributes 1 term of linear transformation $\Lambda$.

## Exercise 2.10

$$T^{i'j'\cdots N'}{}_{k'\ell'\cdots m'} = T^{ij\cdots N}{}_{k\ell\cdots m}\Lambda^{i'}{}_{i}\Lambda^{j'}{}_{j}\cdots\Lambda^{N'}{}_{N}\Lambda^{k}{}_{k'}\Lambda^{\ell}{}_{\ell'}\cdots\Lambda^{m}{}_{m'}.$$

If $T^{ij\cdots N}{}_{k\ell\cdots m}$ are all zero then no linear transformation from above can make $T^{i'j'\cdots N'}{}_{k'\ell'\cdots m'}$ non-zero, since $0\cdot\Lambda^{i}{}_{j}=0$.

$\Rightarrow$ If all components are zero in one basis they are zero in all basis.

## Exercise 2.11

$\{A^{i}{}_{j},\ i,j=1,\ldots,n\}$, $n$-dim vector space.

We define the transformation to be

$$A^{i'}{}_{k'} = \Lambda^{i'}{}_{j}\Lambda^{\ell}{}_{k'}A^{j}{}_{\ell}$$

such that these are components of $A$ on new basis $\{\vec{e}_{j'}\}$. Show that this $A$ is a tensor.

Tensor: tensor of type $\binom{N}{M}$ at $P$ is defined to be a linear function which takes as arguments $M$ vectors and $N$ one-forms and whose value is a real number.

$A^{i}{}_{j}$ transforms same as components of $\binom{N}{M}$-tensor as defined in Exercise 2.9, so the transformation under change of basis is not an obstacle to be tensor for $A^{i}{}_{j}$.

Now we only need to show the mapping $A(\tilde{\omega},\vec{V})$ is a real number for it to be tensor.

We only know the components and that $A^{i}{}_{j}$ is in $n$-dim vector space such that in associated basis $\{\vec{e}_{i}\}$ the components are $A^{i}{}_{j}$, while in $\{\vec{e}_{j'}\}$

$$A^{i'}{}_{j'} = \Lambda^{i'}{}_{j}\Lambda^{k}{}_{j'}A^{j}{}_{k}.$$

The basis have relation

$$\vec{e}_{j'} = \Lambda^{i}{}_{j'}\vec{e}_{i},$$

meaning the transformation

$$A^{i'}{}_{j'} = \Lambda^{i'}{}_{j}\Lambda^{k}{}_{j'}A^{j}{}_{k}$$

associates with one basis vector and one one-form

$$\tilde{\omega}^{j'} = \Lambda^{j'}{}_{\ell}\tilde{\omega}^{\ell}.$$

$\Rightarrow$ We can write $A$ in terms of outer products of $\vec{e}_{i}$ and $\tilde{\omega}^{j}$.

$$\Rightarrow \quad A = A^{i}{}_{j}\tilde{\omega}^{j}\otimes\vec{e}_{i}.$$

This is a tensor as it is a linear function which takes as argument 1 vector & 1 one-form and gives real number, so both definitions define same tensor. $\square$

## Exercise 2.12

Find matrices $\Lambda$ which cast the following matrices into their unit diagonal form.

### a)

Let's use eigenvalues to find $\Lambda$. $\det(A-\lambda I)=0 \leftarrow$ eigenvalue eq.

$$A=\begin{pmatrix}2&1\\1&2\end{pmatrix} \quad\Rightarrow\quad \det(A-\lambda I)=\begin{vmatrix}2-\lambda&1\\1&2-\lambda\end{vmatrix}=(2-\lambda)^2-1=0.$$

$$\Rightarrow \quad \lambda=1, \quad \lambda=3. \quad \text{So we can find } Q^{T}AQ=\begin{pmatrix}3&0\\0&1\end{pmatrix}.$$

Using the spectral theorem: Every real symmetric matrix $A = A^T$ can be diagonalized using an orthogonal matrix $Q$ such that

$$Q^T A Q = D, \qquad \text{where } Q^T Q = I, \quad D \text{ is diagonal,}$$

with $D = \text{diag}(\lambda_1, \ldots, \lambda_n)$ and $Q$ is composed from orthogonal eigenvectors.

Let's find the eigenvectors of $A$ using $Av = \lambda v$.

$$(A - \lambda I)v = 0$$

$$\Rightarrow \quad \left[\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} - \begin{pmatrix} 3 & 0 \\ 0 & 3 \end{pmatrix}\right] \begin{pmatrix} x \\ y \end{pmatrix} = 0$$

$$\Rightarrow \quad \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \quad \Rightarrow \quad x = y.$$

$$\Rightarrow \quad v_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix} \quad \text{for } \lambda_1 = 3.$$

$$(A - \lambda_2 I)v = 0, \quad \begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \quad \Rightarrow \quad x = -y.$$

$$\Rightarrow \quad v_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -1 \end{pmatrix} \quad \text{for } \lambda_2 = 1.$$

$$\Rightarrow \quad Q = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad \Rightarrow \quad Q^T A Q = D = \begin{pmatrix} 3 & 0 \\ 0 & 1 \end{pmatrix}.$$

To get identity we need to rescale $v_1$. Let's define

$$H = \begin{pmatrix} 1/\sqrt{3} & 0 \\ 0 & 1 \end{pmatrix}.$$

$\Rightarrow$ $\Lambda = QH$ gets us the unit diagonal form.

$$\Lambda = \frac{1}{\sqrt{2}} \begin{pmatrix} 1/\sqrt{3} & 1 \\ 1/\sqrt{3} & -1 \end{pmatrix}, \qquad \Lambda^T = \frac{1}{\sqrt{2}} \begin{pmatrix} 1/\sqrt{3} & 1/\sqrt{3} \\ 1 & -1 \end{pmatrix}.$$

$$\begin{aligned} \Lambda^T A \Lambda &= \frac{1}{2} \begin{pmatrix} 1/\sqrt{3} & 1/\sqrt{3} \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix} \begin{pmatrix} 1/\sqrt{3} & 1 \\ 1/\sqrt{3} & -1 \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} (2+1)/\sqrt{3} & (1+2)/\sqrt{3} \\ 1 & -1 \end{pmatrix} \begin{pmatrix} 1/\sqrt{3} & 1 \\ 1/\sqrt{3} & -1 \end{pmatrix} \\ &= \frac{1}{2} \begin{pmatrix} \frac{1}{3}(3+3) & 0 \\ 0 & 2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I. \end{aligned}$$

So $\Lambda = \frac{1}{\sqrt{2}} \begin{pmatrix} 1/\sqrt{3} & 1 \\ 1/\sqrt{3} & -1 \end{pmatrix}$ casts $\begin{pmatrix} 2 & 1 \\ 1 & 2 \end{pmatrix}$ to its unit diagonal form.

**b)**

$$A = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}, \qquad \text{eigenvalues: } \begin{vmatrix} \lambda & 1 \\ 1 & \lambda \end{vmatrix} = \lambda^2 - 1 = 0 \quad \Rightarrow \quad \lambda = \pm 1.$$

$$\lambda_1 = 1 \quad \Rightarrow \quad (A - \lambda_1 I)v_1 = 0,$$

$$\begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix} = 0 \quad \Rightarrow \quad x = y \quad \Rightarrow \quad v_1 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ 1 \end{pmatrix}.$$

$$\lambda_2 = -1 \quad \Rightarrow \quad (A - \lambda_2 I)v_2 = 0,$$
$$\begin{pmatrix} 1 & 1 \\ 1 & 1 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 0 \quad \Rightarrow \quad x = -y \quad \Rightarrow \quad v_2 = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ -1 \end{pmatrix}.$$
$$\Rightarrow \quad Q = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} \quad \text{for } D = \text{diag}(1, -1).$$

We get $\Lambda$ by rescaling $v_2$: $H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$.

$$\Rightarrow \quad \Lambda = QH = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix}.$$

$$\begin{aligned}\Lambda^T A \Lambda &= \frac{1}{2}\begin{pmatrix} 1 & 1 \\ -1 & 1 \end{pmatrix}\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} \\ &= \frac{1}{2}\begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}\begin{pmatrix} 1 & -1 \\ 1 & 1 \end{pmatrix} = \frac{1}{2}\begin{pmatrix} 2 & 0 \\ 0 & -2 \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}.\end{aligned}$$

This is the unit diagonal of $\begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$.

Note: we don't actually need $H = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$ since we cannot use it to change the sign of unit diagonal, so this works with just $\Lambda = Q$.

**c)**

$$A = \begin{pmatrix} 4 & 0 \\ 0 & -1 \end{pmatrix} \quad \Rightarrow \quad \text{eigenvalues: } 0 = \begin{vmatrix} 4 - \lambda & 0 \\ 0 & -1 - \lambda \end{vmatrix}$$
$$\Rightarrow \quad -(4 - \lambda)(1 + \lambda) = 0 \quad \Rightarrow \quad \lambda_1 = 4, \quad \lambda_2 = -1.$$

$$\lambda_1 = 4 \quad \Rightarrow \quad (A - \lambda_1 I)v_1 = \begin{pmatrix} 0 & 0 \\ 0 & -5 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 0$$
$$\Rightarrow \quad y = 0, \quad x = \text{free} \quad \Rightarrow \quad v_1 = \begin{pmatrix} 1 \\ 0 \end{pmatrix},$$
$$\lambda_2 = -1 \quad \Rightarrow \quad (A - \lambda_2 I)v_2 = \begin{pmatrix} 5 & 0 \\ 0 & 0 \end{pmatrix}\begin{pmatrix} x \\ y \end{pmatrix} = 0$$
$$\Rightarrow \quad x = 0, \quad y = \text{free} \quad \Rightarrow \quad v_2 = \begin{pmatrix} 0 \\ 1 \end{pmatrix}.$$
$$\Rightarrow \quad Q = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I \quad \text{for } D = \begin{pmatrix} 4 & 0 \\ 0 & -1 \end{pmatrix}.$$

To get unit diagonal we rescale:

$$H = \begin{pmatrix} 1/\sqrt{4} & 0 \\ 0 & 1 \end{pmatrix} \quad \Rightarrow \quad \Lambda = QH = H = \begin{pmatrix} 1/\sqrt{4} & 0 \\ 0 & 1 \end{pmatrix}.$$

$$\begin{aligned}\Lambda^T A \Lambda &= \begin{pmatrix} 1/\sqrt{4} & 0 \\ 0 & 1 \end{pmatrix}\begin{pmatrix} 4 & 0 \\ 0 & -1 \end{pmatrix}\begin{pmatrix} 1/\sqrt{4} & 0 \\ 0 & 1 \end{pmatrix} \\ &= \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} = I.\end{aligned}$$

So $\Lambda = \begin{pmatrix} 1/\sqrt{4} & 0 \\ 0 & 1 \end{pmatrix}$. For this we should've just recognized that $A$ was already diagonal lol.

## Exercise 2.13

### a)

$$g(\vec{U},\vec{V}) = \vec{U}\cdot\vec{V} \in \mathbb{R} \quad \Rightarrow \quad g \text{ is a } \binom{0}{2} \text{ tensor.}$$

Let's look at how $g^{-1}$'s components transform one-forms by looking at components:

$$g^{-1}(\tilde{h},\tilde{x}) = g^{ij}h_i x_j \in \mathbb{R} \quad \Rightarrow \quad g^{-1} \text{ maps 2 one-forms to real number.}$$

$\Rightarrow$ $g^{-1}$ is bilinear map of one-forms with components $g^{ij}$.

### b)

$$g(\vec{e}_i,\vec{e}_j) = g_{ij} = \pm\delta_{ij}, \qquad \text{since } \{\vec{e}_i\} \text{ form orthonormal basis.}$$

We can write $g$ as $g = g_{ij}\tilde{\omega}^i \otimes \tilde{\omega}^j$. Dual basis: $\tilde{\omega}^i(\vec{e}_j) = \delta^i_j$.

$$\begin{aligned} g(\vec{e}_i,\vec{e}_j) &= g_{\ell k}\tilde{\omega}^\ell(\vec{e}_i)\tilde{\omega}^k(\vec{e}_j) \\ &= g_{\ell k}\delta^\ell_i\delta^k_j, \qquad \text{since } g \text{ is canonical in this basis } g_{\ell k} = \pm\delta_{\ell k} \\ &= \pm\delta_{ij}. \end{aligned}$$

Similarly due to $g$ being canonical so is $g^{-1}$ with only diagonal entries being non-zero and $\pm 1$.

$$\Rightarrow \quad g^{-1} = g^{ij}\vec{e}_i \otimes \vec{e}_j \quad \text{is inverse of } g.$$

$$\begin{aligned} g^{-1}(\tilde{\omega}^i,\tilde{\omega}^j) &= g^{\ell k}\vec{e}_\ell(\tilde{\omega}^i)\vec{e}_k(\tilde{\omega}^j) \\ &= (\pm\delta^{\ell k})\delta^i_\ell\delta^j_k = \pm\delta^{ij}. \quad \square \end{aligned}$$

$\Rightarrow$ $\{\tilde{\omega}^i\}$ is an orthonormal basis.

## Exercise 2.14

Let's start from general diagonal metric $g$ (since we can always diagonalise $g$).

$$g = \text{diag}(f_1(P),\ldots,f_n(P)) = \text{diag}(f_1,\ldots,f_n)$$

for some $C^\infty$ function $f$ of $P$.

We can do the transformation at point $P$ to find a basis where $g$ is canonical. In this case

$$\Lambda = \text{diag}(1/\sqrt{f_1},\ldots,1/\sqrt{f_n})$$

giving

$$g' = \Lambda^T g \Lambda = \text{diag}(\pm 1,\ldots,\pm 1),$$

where the sign of each entry depends on original $g$.

$$\text{(i)} \quad \Rightarrow \quad g'_{ij} = \pm\delta_{ij}. \quad \square$$

(So the basis where $g \to g'$ at $P$ is orthonormal.)

(ii)-condition:

$$\left.\frac{\partial g'_{ij}}{\partial x_k}\right|_P = \left.\frac{\partial}{\partial x_k}\left\{\frac{1}{\sqrt{f}}f\frac{1}{\sqrt{f}}\right\}\right|_P, \qquad \text{for each } f \text{ in } g.$$

From now on we use $x^k$ instead of $x_k$.

$$\left.\frac{\partial g'_{ij}}{\partial x^k}\right|_P = \left[-\frac{1}{2}f^{-3/2}\frac{\partial f}{\partial x^k}ff^{-1/2} + f^{-1/2}\frac{\partial f}{\partial x^k}f^{-1/2} - \frac{1}{2}f^{-1/2}f\frac{\partial f}{\partial x^k}f^{-3/2}\right]_P$$
$$= \left[-f^{-1}\frac{\partial f}{\partial x^k} + f^{-1}\frac{\partial f}{\partial x^k}\right]_P = 0. \quad \square$$

For (iii)-condition we just continue with additional derivative.

$$\left.\frac{\partial^2 g'_{ij}}{\partial x^k \partial x^\ell}\right|_P = \left.\frac{\partial^2}{\partial x^k \partial x^\ell}\{f^{-1/2}ff^{-1/2}\}\right|_P$$
$$= \left[\left(\frac{\partial^2 f^{-1/2}}{\partial x^k \partial x^\ell}\right)ff^{-1/2} + f^{-1/2}\left(\frac{\partial^2 f}{\partial x^k \partial x^\ell}\right)f^{-1/2}\right.$$
$$\left. + f^{-1/2}f\left(\frac{\partial^2 f^{-1/2}}{\partial x^k \partial x^\ell}\right)\right]_P,$$
$$\left.\frac{\partial^2 g'_{ij}}{\partial x^k \partial x^\ell}\right|_P = \left[2f^{1/2}\left(\frac{\partial^2 f^{-1/2}}{\partial x^k \partial x^\ell}\right) + f^{-1}\frac{\partial^2 f}{\partial x^k \partial x^\ell}\right]_P.$$

This can be non-zero $\Rightarrow$ Not necessarily zero. $\square$
$\Rightarrow$ a $C^\infty$ metric tensor is locally flat.

## Exercise 2.15

Euclidean metric is $g = \mathrm{diag}(1,1)$ in Cartesian coords with basis $(\partial/\partial x, \partial/\partial y)$.

### a)

New basis $\partial/\partial r$, $\partial/\partial\theta$.

$$x = r\cos\theta, \qquad y = r\sin\theta,$$
$$r = \sqrt{x^2+y^2}, \qquad \tan\theta = \frac{y}{x} \quad \Rightarrow \quad \theta = \arctan\left(\frac{y}{x}\right), \qquad 0 \le \theta \le \pi.$$

Vector transformation under coordinate change:

$$\frac{\partial}{\partial r} = \frac{\partial x^i}{\partial r}\frac{\partial}{\partial x^i}, \qquad \frac{\partial}{\partial \theta} = \frac{\partial x^i}{\partial \theta}\frac{\partial}{\partial x^i}.$$

$$\Rightarrow \quad \Lambda^{i'}{}_j = \frac{\partial x^{i'}}{\partial x^j}, \qquad \text{where } x^{i'} = \{r,\theta\}, \text{ i.e., primed are new coordinates.}$$

The inverse matrix of this is

$$\Lambda^i{}_{j'} = \frac{\partial x^i}{\partial x^{j'}}, \qquad x^{j'} = \{r,\theta\}.$$

The metric transforms as:

$$g_{i'j'} = \Lambda^i{}_{i'}\Lambda^j{}_{j'}g_{ij}.$$

$$g_{rr} = \frac{\partial x^i}{\partial r}\frac{\partial x^j}{\partial r}g_{ij} = \left(\frac{\partial x}{\partial r}\right)^2 g_{11} + \left(\frac{\partial y}{\partial r}\right)^2 g_{22}$$
$$= \cos^2\theta + \sin^2\theta = 1.$$

$$\frac{\partial x}{\partial r} = \left(\frac{\partial r}{\partial x}\right)^{-1} = \cos\theta, \qquad \frac{\partial y}{\partial r} = \left(\frac{1}{2}\frac{1}{r}2y\right)^{-1} = \sin\theta.$$

$$
\begin{aligned}
g_{\theta r} = g_{r\theta} &= \frac{\partial x^i}{\partial r}\frac{\partial x^j}{\partial \theta} g_{ij} \\
&= \frac{\partial x}{\partial r}\frac{\partial x}{\partial \theta} + \frac{\partial y}{\partial r}\frac{\partial y}{\partial \theta} \\
&= -r\cos\theta\sin\theta + r\sin\theta\cos\theta = 0,
\end{aligned}
$$

with $\partial x/\partial\theta = -r\sin\theta$, $\partial y/\partial\theta = r\cos\theta$.

$$
\begin{aligned}
g_{\theta\theta} &= \frac{\partial x^i}{\partial \theta}\frac{\partial x^j}{\partial \theta} g_{ij} \\
&= \left(\frac{\partial x}{\partial \theta}\right)^2 g_{11} + \left(\frac{\partial y}{\partial \theta}\right)^2 g_{22} \\
&= r^2(\sin^2\theta + \cos^2\theta) = r^2.
\end{aligned}
$$

$$
\Rightarrow \quad g = \begin{pmatrix} 1 & 0 \\ 0 & r^2 \end{pmatrix}
$$

is the Euclidean metric in polar coords $(\partial/\partial r, \partial/\partial\theta)$.

**b)**

$$
\hat{r} = \cos\theta\,\hat{x} + \sin\theta\,\hat{y}, \qquad \hat{\theta} = -\sin\theta\,\hat{x} + \cos\theta\,\hat{y},
$$

with $\hat{x} = \partial/\partial x$, $\hat{y} = \partial/\partial y$.

$$
\begin{aligned}
\hat{r} &= \cos\theta\frac{\partial}{\partial x} + \sin\theta\frac{\partial}{\partial y}, \\
\hat{\theta} &= -\sin\theta\frac{\partial}{\partial x} + \cos\theta\frac{\partial}{\partial y}.
\end{aligned}
$$

Comparing with the basis from (a):

$$
\begin{aligned}
&\frac{\partial}{\partial r} = \frac{\partial x^i}{\partial r}\frac{\partial}{\partial x^i} = \frac{\partial x}{\partial r}\frac{\partial}{\partial x} + \frac{\partial y}{\partial r}\frac{\partial}{\partial y}, \qquad \frac{\partial x}{\partial r} = \cos\theta, \\
&\frac{\partial}{\partial r} = \cos\theta\frac{\partial}{\partial x} + \sin\theta\frac{\partial}{\partial y} \quad \Rightarrow \quad \hat{r} = \partial/\partial r, \\
&\frac{\partial}{\partial \theta} = \frac{\partial x}{\partial \theta}\frac{\partial}{\partial x} + \frac{\partial y}{\partial \theta}\frac{\partial}{\partial y}, \qquad \frac{\partial x}{\partial \theta} = -r\sin\theta, \\
&\quad = -r\sin\theta\frac{\partial}{\partial x} + r\cos\theta\frac{\partial}{\partial y} \quad \Rightarrow \quad \hat{\theta} = \frac{1}{r}\frac{\partial}{\partial \theta}.
\end{aligned}
$$

The coordinates $\hat{r}, \hat{\theta}$ in terms of $\partial/\partial r$ & $\partial/\partial\theta$ are

$$
\begin{cases}
\hat{r} = \partial/\partial r, \\
\hat{\theta} = \dfrac{1}{r}\,\partial/\partial\theta.
\end{cases}
$$

Now let's get $g$ in $(\hat{r}, \hat{\theta})$ basis.

$$
\begin{aligned}
g_{\hat{r}\hat{r}} &= g(\hat{r},\hat{r}) = g(\cos\theta\,\hat{x} + \sin\theta\,\hat{y}, \cos\theta\,\hat{x} + \sin\theta\,\hat{y}) \\
&= \cos^2\theta\,\underbrace{g(\hat{x},\hat{x})}_{=1} + 2\sin\theta\cos\theta\,\underbrace{g(\hat{x},\hat{y})}_{=0} + \sin^2\theta\,\underbrace{g(\hat{y},\hat{y})}_{=1}, \\
g_{\hat{r}\hat{r}} &= 1.
\end{aligned}
$$

Here we used the fact that $g$ is bilinear function of two vectors. Similarly

$$\begin{aligned} g_{\hat\theta\hat r} = g_{\hat r\hat\theta} &= g(\cos\theta\,\hat x + \sin\theta\,\hat y, -\sin\theta\,\hat x + \cos\theta\,\hat y) \\ &= -\cos\theta\sin\theta\,\underbrace{g(\hat x,\hat x)}_{=1} + \cos\theta\sin\theta\,\underbrace{g(\hat y,\hat y)}_{=1} + \text{terms}\cdot\underbrace{g(\hat x,\hat y)}_{=0} \\ &= 0, \\ g_{\hat\theta\hat\theta} &= g(-\sin\theta\,\hat x + \cos\theta\,\hat y, -\sin\theta\,\hat x + \cos\theta\,\hat y) \\ &= \sin^2\theta\,\underbrace{g(\hat x,\hat x)}_{=1} + \cos^2\theta\,\underbrace{g(\hat y,\hat y)}_{=1} + \text{terms}\cdot\underbrace{g(\hat x,\hat y)}_{=0} \\ &= 1. \end{aligned}$$

$\Rightarrow$ in the $\{\hat r, \hat\theta\}$ basis the metric $g$ is

$$g = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

## Exercise 2.16

$df$ in the coordinate basis $(\partial/\partial x, \partial/\partial y)$ has the form

$$\widetilde{df} = \left(\frac{\partial f}{\partial x}, \frac{\partial f}{\partial y}\right) \quad \Rightarrow \quad \widetilde{df}_i = \frac{\partial f}{\partial x^i}.$$

One-forms transform by $\Lambda^i{}_{i'}$, the inverse matrix of the one for vector transformation:

$$\Lambda^i{}_{i'} = \frac{\partial x^i}{\partial x^{i'}}, \qquad \text{where primed coords are the new basis.}$$

$$\Rightarrow \quad \widetilde{df}_{i'} = \Lambda^i{}_{i'}\frac{\partial f}{\partial x^i} = \frac{\partial x^i}{\partial x^{i'}}\frac{\partial f}{\partial x^i}.$$

$$\begin{aligned} \widetilde{df}_r &= \frac{\partial x}{\partial r}\frac{\partial f}{\partial x} + \frac{\partial y}{\partial r}\frac{\partial f}{\partial y} = 2\frac{\partial f}{\partial r}, \qquad \text{by chain rule,} \\ \widetilde{df}_\theta &= \frac{\partial x}{\partial \theta}\frac{\partial f}{\partial x} + \frac{\partial y}{\partial \theta}\frac{\partial f}{\partial y} = 2\frac{\partial f}{\partial \theta}. \end{aligned}$$

$$\Rightarrow \quad \widetilde{df} = \left(\frac{\partial f}{\partial r}, \frac{\partial f}{\partial \theta}\right), \qquad \text{this makes sense since we're in coordinate basis.}$$

As for $\overrightarrow{df}$ we can just use $g$ in polar coords:

$$g = \begin{pmatrix} 1 & 0 \\ 0 & r^2 \end{pmatrix} \quad \Rightarrow \quad (df)^{i'} = g^{i'j'}\widetilde{df}_{j'}, \qquad (g)^{-1} = \begin{pmatrix} 1 & 0 \\ 0 & r^{-2} \end{pmatrix}.$$

$$\overrightarrow{df} = \left(\widetilde{df}_1, \frac{1}{r^2}\widetilde{df}_2\right) = \left(\frac{\partial f}{\partial r}, r^{-2}\frac{\partial f}{\partial \theta}\right).$$

Now let's go to non-coordinate basis $(\hat r, \hat\theta)$.

$$\hat r = \cos\theta\,\hat x + \sin\theta\,\hat y, \qquad \hat\theta = -\sin\theta\,\hat x + \cos\theta\,\hat y,$$

$$\hat r = \partial/\partial r, \qquad \hat\theta = \frac{1}{r}\,\partial/\partial\theta.$$

$$\Rightarrow \quad \widetilde{df}(\hat x) = \frac{\partial f}{\partial x}, \qquad \widetilde{df}(\hat y) = \frac{\partial f}{\partial y}.$$

In same vein:

$$\widetilde{df}\left(\frac{\partial}{\partial r}\right) = \frac{\partial f}{\partial r}, \qquad \widetilde{df}\left(\frac{\partial}{\partial \theta}\right) = \frac{\partial f}{\partial \theta}.$$

$$\begin{aligned}
\widetilde{df}(\hat{r}) &= \widetilde{df}(\cos\theta\,\hat{x} + \sin\theta\,\hat{y}) \\
&= \cos\theta\,\widetilde{df}(\hat{x}) + \sin\theta\,\widetilde{df}(\hat{y}) \\
&= \cos\theta\frac{\partial f}{\partial x} + \sin\theta\frac{\partial f}{\partial y} = \left(\cos\theta\frac{\partial}{\partial x} + \sin\theta\frac{\partial}{\partial y}\right) f.
\end{aligned}$$

But we can also write this in polar coords $(r, \theta)$:

$$\widetilde{df}(\hat{r}) = \widetilde{df}(\partial/\partial r) = \frac{\partial f}{\partial r}, \qquad \text{(stays same which is expected)},$$

$$\widetilde{df}(\hat{\theta}) = \widetilde{df}\left(\frac{1}{r}\frac{\partial}{\partial \theta}\right) = \frac{1}{r}\frac{\partial f}{\partial \theta}, \qquad \text{(rescaled by } 1/r\text{, expected)}.$$

$$\Rightarrow \quad \widetilde{df} = \left(\frac{\partial f}{\partial r}, \frac{1}{r}\frac{\partial f}{\partial \theta}\right) \quad \text{in } \{\hat{r}, \hat{\theta}\} \text{ basis.}$$

$$\Rightarrow \quad (df)^j = g^{ij}\widetilde{df}_i = \left(\frac{\partial f}{\partial r}, \frac{1}{r}\frac{\partial f}{\partial \theta}\right),$$

where $g^{ij}$ here is for $g^{-1}$ in $(\hat{r}, \hat{\theta})$ basis which is $I$.